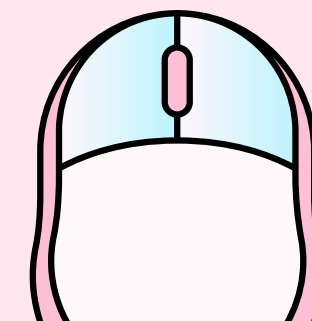
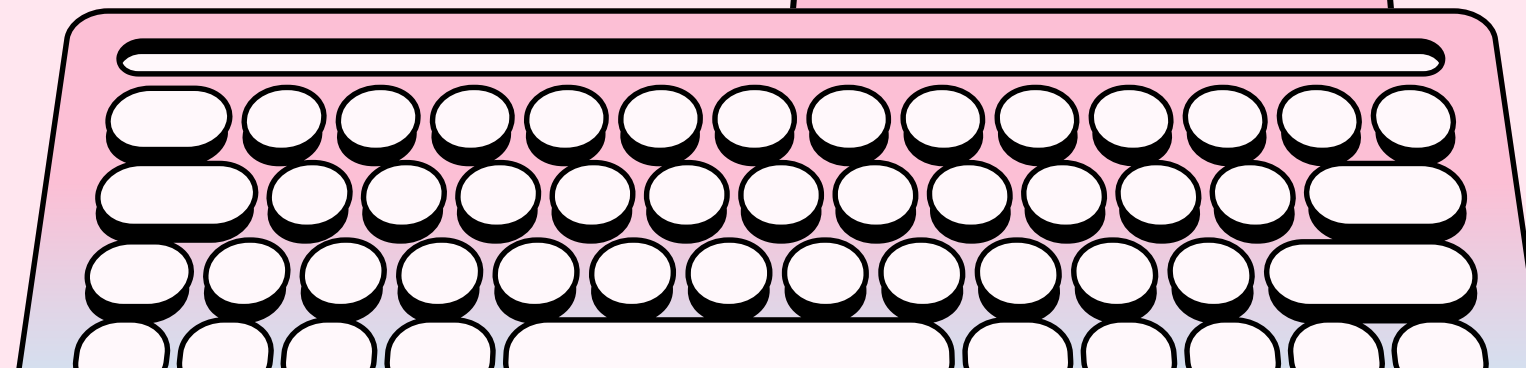
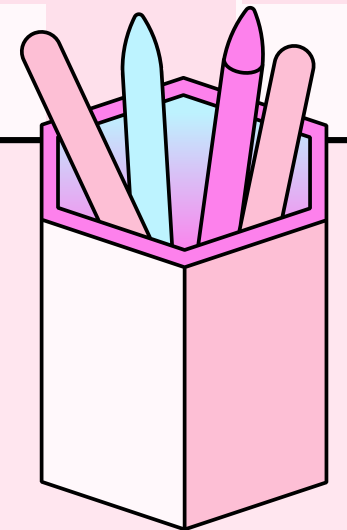
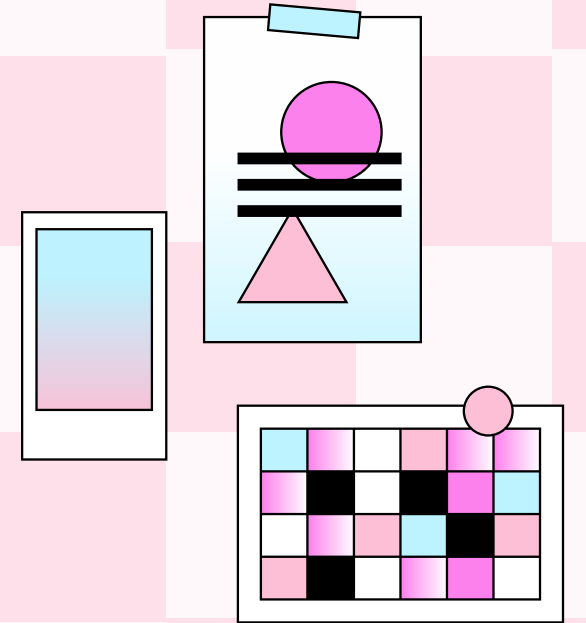
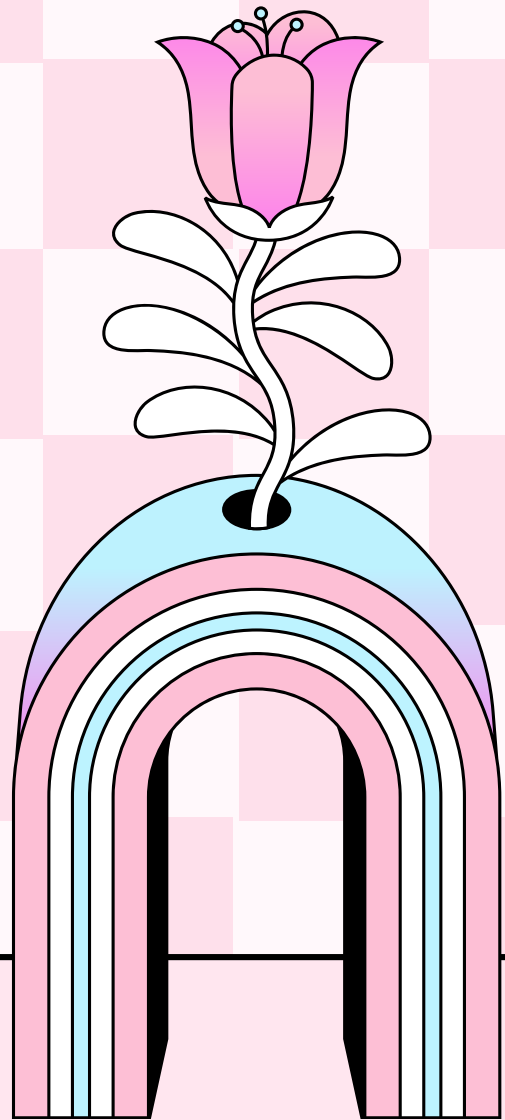


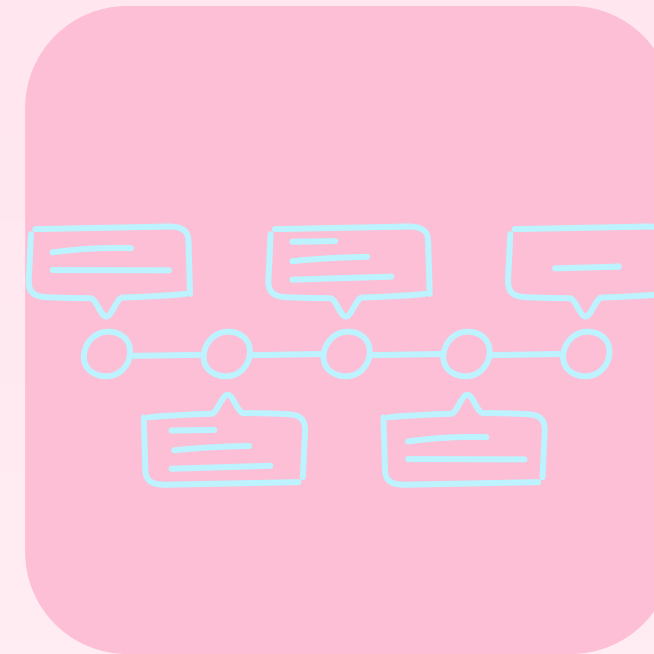
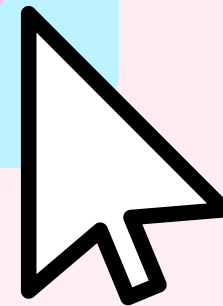
3.1.2 Product Lifecycle

HARSHITA AND ETHAN





ORIGINAL



TIMELINE

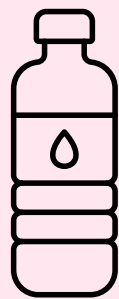


IMPROVEMENTS



THE END

RAW MATERIALS



PLASTIC (DASANI)

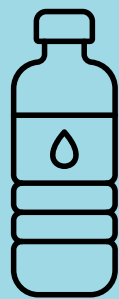
Raw material	Points
Crude oil (for PET plastic bottle and cap)	1
Bauxite (for label lining)	1



METAL (OWALA)

Raw material	Points
Crude oil (for plastic parts)	1
Iron ore (for stainless steel)	1
Natural rubber (or petroleum-based for rubber parts)	1
Silica (for silicone)	1
Bauxite	1
Limestone (used in some coatings or processing)	1

MATERIALS PROCESSING



PLASTIC (DASANI)

Metals or Plastics in Product	Points
PET plastic (bottle)	1
HDPE or PP plastic (cap)	1
Plastic label (polypropylene or polyethylene)	1



METAL (OWALA)

Metals or Plastics in Product	Points
Stainless steel	1
Polypropylene (plastic)	1
Silicone	1
Nylon	1
Rubber	1
Powder coating or paint	1

MANUFACTURING



PLASTIC (DASANI)

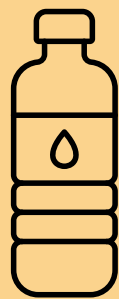
Parts and Peices in Product	Points
Plastic bottle	1
Plastic cap	1
Plastic label	1
Plastic seal	1



METAL (OWALA)

Parts and Peices in Product	Points
Main stainless steel body	1
Plastic lid	1
Flip straw or sipper	1
Silicone gasket/seal	1
Plastic or rubber grip	1
Carrying handle	1

PACKAGING



PLASTIC (DASANI)

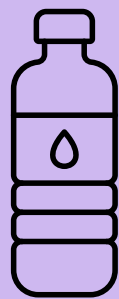
Parts and Peices in Product	Points
Plastic Only	15



METAL (OWALA)

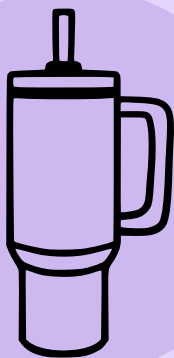
Parts and Peices in Product	Points
Plastic and cardboard	10
Instruction sheet included	5

DISTRIBUTION (TRANSPORTATION)



PLASTIC (DASANI)

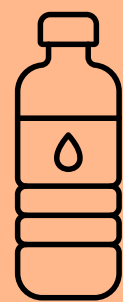
Transportation	Points
Yes (Plane, truck, and boat)	15



METAL (OWALA)

Transportation	Points
Yes (Plane, truck, and boat)	15

UTILIZATION



PLASTIC (DASANI)

Use of Product	Points
Can only be used once	15

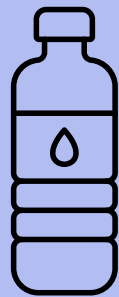


METAL (OWALA)

Use of Product	Points
Product can be used for over 10 years	5

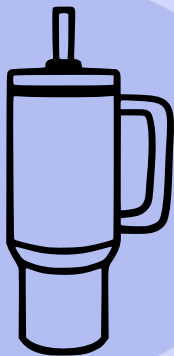


DISPOSAL



PLASTIC (DASANI)

Parts and Peices in Product	Points
All of the product can be recycled	0



METAL (OWALA)

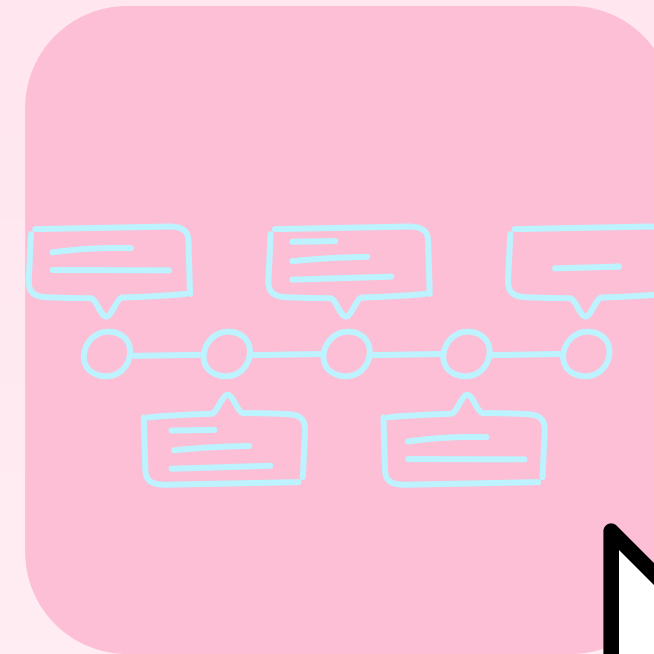
Parts and Peices in Product	Points
Some product materials can be recycled (like stainless steel and some plastics)	5

OVERALL/COMPARISON

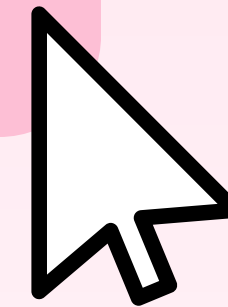
Category	 Dasani (Plastic)	 Owala (Metal)
Total Score	54	53
Raw Materials	Fewer types, lower score	More types, higher score
Processing	Simple plastics	Mixed materials
Manufacturing	Few parts	Complex build
Packaging	All plastic	Paper + plastic = equal
Transport	Long-distance	Same as Dasani
Use	Single-use = high score	Reusable = low score
Disposal	Fully recyclable	Partly recyclable



ORIGINAL



TIMELINE



IMPROVEMENTS



THE END

DASANI

X

OWALA

X

PLASTIC (DASANI) TIMELINE

RAW MATERIAL EXTRACTION

PETROLEUM EXTRACTED FOR PET PLASTIC

MATERIALS PROCESSING

PET POLYMERIZED AND SHAPED

MANUFACTURING

BOTTLE MOLDED, FILLED, CAPPED,
LABELED

PACKAGING

SHRINK-WRAPPED IN PLASTIC

DISTRIBUTION

TRUCKED TO STORES

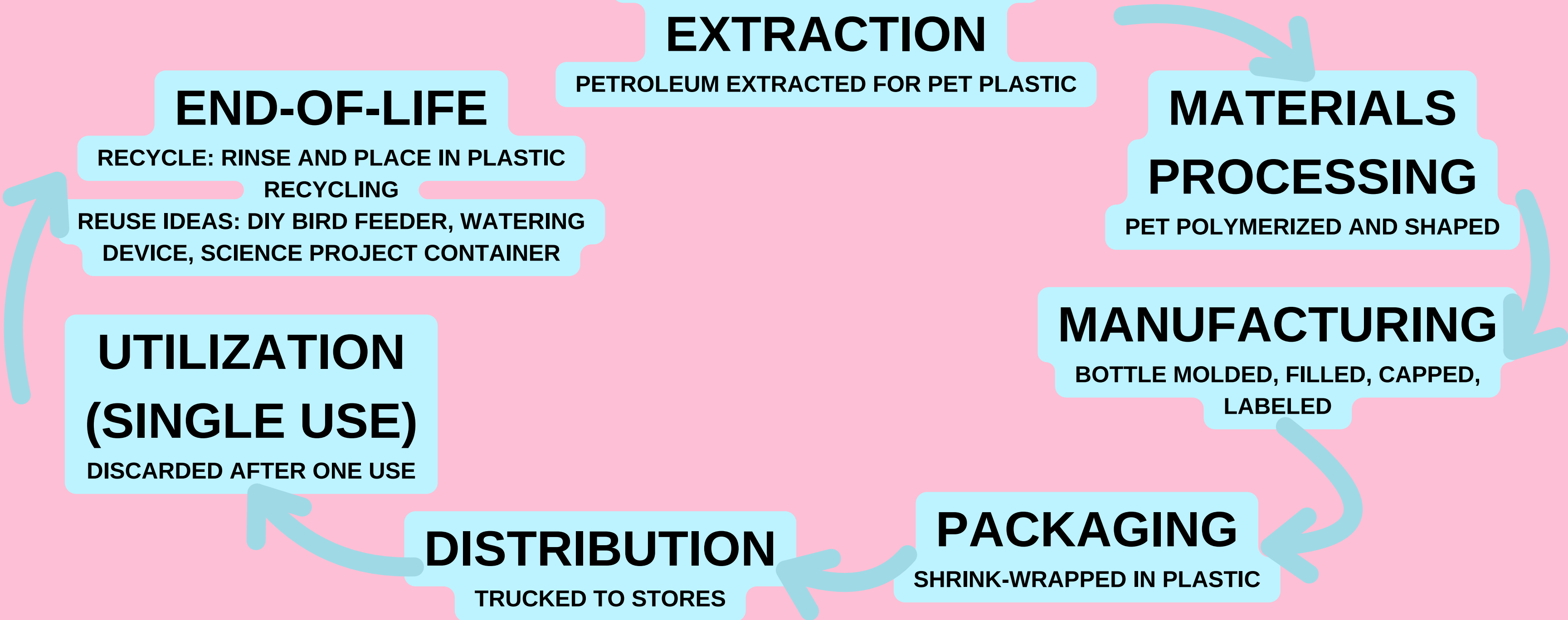
UTILIZATION (SINGLE USE)

DISCARDED AFTER ONE USE

END-OF-LIFE

RECYCLE: RINSE AND PLACE IN PLASTIC
RECYCLING

REUSE IDEAS: DIY BIRD FEEDER, WATERING
DEVICE, SCIENCE PROJECT CONTAINER



DASANI

X

OWALA

X

MÉTAL (OWALA) TIMELINE

RAW MATERIAL EXTRACTION

MINING IRON ORE (FOR STAINLESS STEEL), EXTRACTING CRUDE OIL (FOR PLASTICS), AND HARVESTING RUBBER/SILICA

END OF LIFE / DISPOSAL

MAY WEAR OUT, BREAK, OR BE REPLACED AFTER A DECADE OF USE.

RECYCLE: SEPARATE MATERIALS BY TYPE
REUSE IDEAS: PLANTER, TOOL HOLDER, PIGGY BANK

UTILIZATION (10+ YEARS)

REFILLED AND REUSED DAILY — REDUCES SINGLE-USE PLASTIC WASTE.

DISTRIBUTION

SHIPPED VIA TRUCK, BOAT, OR PLANE
TO RETAILERS

PACKAGING

PLACED IN RECYCLABLE CARDBOARD AND
MINIMAL PLASTIC WRAP.

MATERIALS PROCESSING

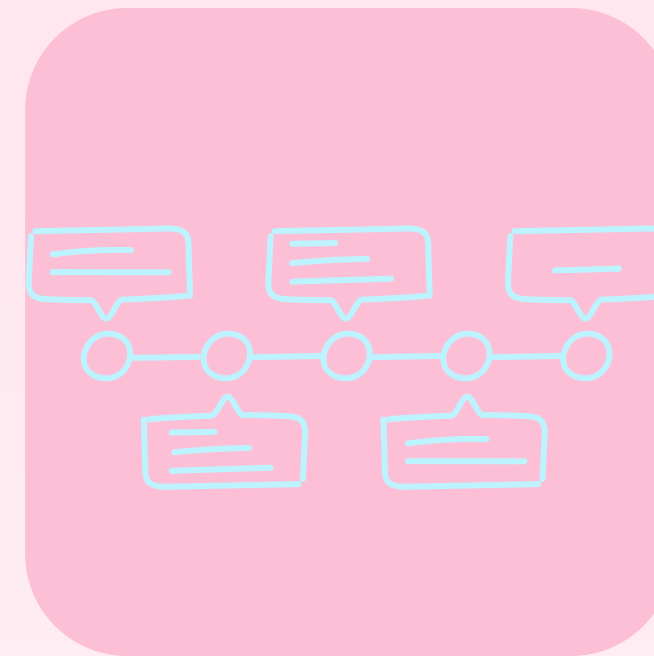
REFINED INTO STAINLESS STEEL,
POLYPROPYLENE, SILICONE, ETC.

MANUFACTURING

MOLDED INTO BOTTLE PARTS (BODY,
CAP, SEALS), THEN ASSEMBLED



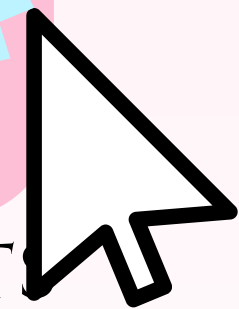
ORIGINAL



TIMELINE



IMPROVEMENTS



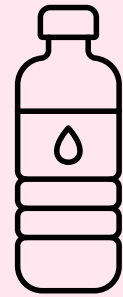
THE END

IMPROVE... X

TRADEO... X

OVERALL... X

IMPROVEMENTS



PLASTIC (DASANI)

- RECYCLED PLASTIC → ♻️ (↓ RAW/PROCESSING)
- THICKER BOTTLE = REUSE (↓ UTILIZATION)
- SLIMMER DESIGN (↓ RAW USE)
- CLEAR RECYCLE LABEL (↓ DISPOSAL)
- **NEW SCORE: 42**



METAL (OWALA)

- RECYCLED METAL & PLASTIC (↓ RAW/PROCESSING)
- ALL-CARDBOARD PACKAGING (↓ PACKAGING)
- FEWER PARTS (↓ MANUFACTURING)
- EASY TO RECYCLE PARTS (↓ DISPOSAL)
- REPLACEABLE PARTS (↓ UTILIZATION)
- LOCAL PRODUCTION (↓ TRANSPORT)
- **NEW SCORE: 30**

IMPROVE... X

TRADEO... X

OVERALL... X

TRADEOFFS TO IMPROVMENTS



PLASTIC (DASANI)

- RECYCLED PLASTIC → MAY COST MORE, LIMITED SUPPLY
- THICKER BOTTLE → HEAVIER, HIGHER TRANSPORT EMISSIONS
- SLIMMER DESIGN → LESS DURABLE, SMALLER VOLUME
- ENCOURAGING RECYCLING → NOT ALL USERS FOLLOW THROUGH



METAL (OWALA)

- RECYCLED METAL/PLASTIC → HARDER TO SOURCE, HIGHER COST
- CARDBOARD-ONLY PACKAGING → LESS WATERPROOF/DURABLE
- FEWER PARTS → MAY REDUCE FEATURES/USABILITY
- SIMPLIFIED DESIGN FOR RECYCLABILITY → DESIGN FLEXIBILITY LIMITED
- REPLACEABLE PARTS → HIGHER UPFRONT DESIGN COST
- LOCAL PRODUCTION → NOT ALWAYS SCALABLE OR AVAILABLE

IMPROVE... X

TRADEO... X

OVERALL... X



OVERALL ANALYSIS



PLASTIC (DASANI)

- LOWER IMPACT BY USING RECYCLED PLASTIC
- DESIGNED FOR REUSE TO REDUCE WASTE
- SLIMMER SHAPE IMPROVES TRANSPORT EFFICIENCY
- ENCOURAGES USER RECYCLING BEHAVIOR
- **SCORE DROP: 54 → 42**

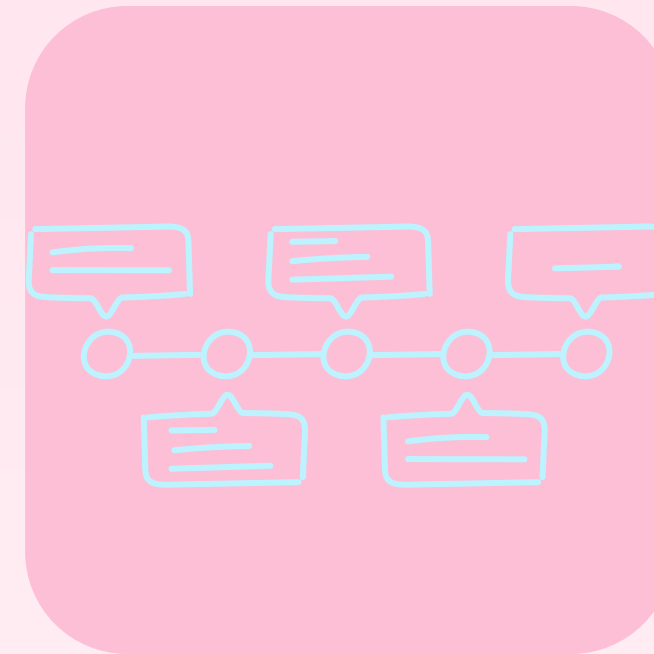


METAL (OWALA)

- USES RECYCLED MATERIALS THROUGHOUT
- SIMPLIFIED PARTS FOR EASIER PRODUCTION
- ALL-CARDBOARD PACKAGING REDUCES WASTE
- DESIGNED FOR REPAIR AND LONG-TERM USE
- BETTER END-OF-LIFE RECYCLABILITY
- **SCORE DROP: 53 → ~30**



ORIGINAL



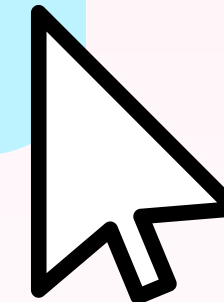
TIMELINE



IMPROVEMENTS



THE END



THANK Y... X

DELIVER... X



THANK YOU!

THANK Y... X

DELIVER... X

DELIVERABLES

1. 1ST INVENTORY ANALYSIS SLIDES 3-9
2. 2ND INVENTORY ANALYSIS SLIDES 3-9
3. COMPARISON SLIDE 10
4. TIMELINE SLIDE 12 AND 13
5. IMPROVEMENT ANALYSIS SLIDE 15-17